

In the past, the software running on a box was very tightly coupled to the hardware box, and thus, it was very difficult to dynamically move an application from one computer to another. However, with the emergence of hypervisor (server virtualisation) technologies like Xen, HyperV, and VMWare, now it is possible to dynamically move applications between computer systems, and thus, increase the overall system utilisation. The hypervisor technology allows multiple applications to run on a single box, and the failure of a single application does not affect the execution of the other applications running on the computer. The hypervisor technology has been around since 1960s (IBM VM operating system), but only now this technology has been made available to run on commodity hardware systems, and thus, it has become more prevalent. Most datacentre operators are re-designing their datacentres to leverage this technology.

Evaluating Resources

Standards organisations like SNIA (Storage Networking Industry Association) are grappling with the task of how to rate a storage device with respect to its power consumption properties. For example, a storage device might use flash technology instead of disk drives, and thus, it can consume less power while being more expensive. So, it is not prudent to just look at the amount of power that is consumed by a computer device in isolation. Instead, one should look at power consumption in conjunction with performance, availability, reliability, physical shelf space requirements, and cost considerations. Therefore, system administrators have to make trade-offs between these different parameters, and there is a need for new power-related metrics like IOs/Watt or IOs/Watt/dollar, or Watts/Cubic Feet etc.

Furthermore, the admins need to look at their respective application workloads to select the proper type of computer resources. For example, there is a difference in the I/O characteristics of archival workloads and online transaction processing (OLTP) type workloads. In archival workloads one does not care about high throughput, whereas, in OLTP workloads throughput requirements are very important. Thus, one can purchase a storage system with slower RPM disks for archival workloads than for OLTP workloads because slower RPM disks consume less power and are usually cheaper.

Ensuring Less Power Consumption

There are both proactive and reactive techniques with respect to reducing power consumption in computer devices.

Proactive techniques: These techniques a priori ensure that devices consume less power. For example, one can cut down on the number of disks being used by using higher capacity disks. A one-terabyte disk will consume less power than ten 100 gigabyte disks. Similarly, one can

reduce the number of copies of data, use data compression, thin provisioning and data de-duplication techniques to reduce the amount of data being stored on disks. This, in turn, reduces the number of disk drives being used which, in turn, leads to less power consumption. Similarly, one can also use Flash drives or higher efficiency power supplies to also proactively cut down on the amount of power being consumed.

Reactive techniques: In reactive techniques one dynamically changes the state of a physical resource from high power consuming state to low power consuming state. The state of CPU, memory and disk drives can be dynamically transitioned between different power states. It is important to note that there is a trade-off between power consumption and performance when one transitions a device to a lower power state. For example, if we spin down or shut down a disk drive, the next time we want to read data from that drive we will incur higher latency. This is not acceptable behavior for all the different types of workloads. For example, interactive applications cannot wait for disks to spin up.

Designing Efficient Cooling

In the past people assumed that for every one watt of power consumed, you requires one watt of power for cooling. However, now people are building sophisticated datacentres to reduce the power required for cooling. Datacentre builders are using the notion of hot aisles and cold aisles, and are also encasing (insulating) the racks to ensure that hot air does not mix with the freshly brought cold air. Datacentre designers are also using blanking panels to fill up empty space in racks in order to manage airflow efficiency. People are also locating datacentres in regions where the outside air temperature and humidity is optimum (temperature range of 20 degree to 25 degree C, and humidity range of 40 to 45 percent with a maximum dew point of 17 degree C). Some system designers have started to leverage water-cooling in lieu of air-cooling in order to more efficiently remove the heat from the hot systems.

However, the plumbing infrastructure requirement for water-cooling leads to higher startup costs. Datacentre designers are also employing raised floor designs to facilitate better airflow circulation. In conclusion, the use of these cooling techniques is now leading to a ratio of less than one watt of cooling for every one watt of power consumed.

In conclusion, it is important to note that in addition to performance, power management is another quantitative way of measuring system performance. Going forward, as standards bodies produce new power measurement units this will become another key differentiator between the products from different vendors. ■

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A STORE LORE

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Many innovations in the IT industry start as complex and expensive technologies aimed at large enterprises, and then later filter down to the small and medium enterprises (SMEs) as simplified, cost-reduced packages. Storage or backup technology is no exception to this rule. Big-ticket concepts such as networked storage, tiered storage and fully-automated and fully-integrated backup, restore and archive functions are now available to SMEs and are being rapidly adopted.

What are the storage requirements of the SME and how do they differ from those of the large enterprise? The three main considerations lie in the area of primary storage performance and expandability, backup and archive function, and staffing costs for installation and maintenance of the data protection system.

Primary Storage

The primary function of the storage system is to reliably store the files and database of the company and allow rapid retrieval by multiple users. For most enterprises, storage choices can be evaluated in terms of availability—that is latency and transfer rate, and cost per megabyte. However, the SME faces a much greater challenge in managing the growth of its storage capacity, which may be rapid but will always be unpredictable. For this reason the SME will look for a storage solution which can start small but can scale easily, with the ability to add storage quickly and easily in small or large modules with minimal disruption to the system while maintaining the same level of accessibility and cost per megabyte. While the large enterprise will plan for a growth in capacity of 20-50 percent per annum, the SME must plan for a growth from zero percent to 200 percent or more per annum.

Backup And Archive

Backup is an essential, integral part of any storage architecture and must be planned and rolled out at the same time as the primary storage. It must also meet the same criteria for cost, performance, reliability and expandability as the primary storage.

Backup covers two major functions, the ability to restore the files and database of the company in the case of catastrophic failure of the primary storage and the ability to make

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copies of essential data for infrequent access for business, regulatory or legal reasons. The usual reasons considered for failure are “earthquake, fire and flood” but in fact the more likely reasons are equipment failure, software failure, viruses and worms, human error and deliberate sabotage. SMEs are more vulnerable to human error and sabotage since equipment is often in an open office environment and is maintained by inexperienced staff.

The archive function sometimes uses the same tape drive as the backup function but recently the trend in companies large and small has been to make frequent backup copies to a separate backup disk device allowing rapid disk-speed backup and restore, and to use tapes only for archival purposes.

It will also be prudent to move backup and archived information on tapes to an off-site remote location to safeguard against local disasters.

Manpower

The third area of consideration is the area of staffing costs which is a particularly difficult area for SMEs. For an organisation of 100 people or so it is difficult to justify more than one full-time IT employee. A larger organisation can have the luxury of having IT staff to manage, and even on-call staff available. For the SME operating 24/7 the worst nightmare is a catastrophic failure requiring restore from backup in the middle of the night or at a weekend when the only IT person is away. Therefore a backup and restore which is fully automated without requiring the presence of skilled staff can improve performance and security and bring down costs in this area.

For the SME the functions of the storage system may best be summarised by the mantra: store-retrieve-backup- restore-archive. Attention should be given to the ability of the storage system to scale easily from very small capacity to very large, with storage increments added easily. It is always better and cheaper to buy and install storage as it is required rather than in advance since prices are always on the downward trend, and performance always improves. The backup and restore architecture must be laid down at the same time as the primary storage and it must also be reliable and scaleable. ■

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